

Durga sai prasad Pragada

Software Engineer (Python & Gen AI)

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PROFESSIONAL SUMMARY

Python & Gen AI Developer with around **2+ years** of experience **building AI-powered backend applications** and **LLM-based solutions**. Proficient in **Python, FastAPI, LangChain**, and **LLM integration**, with hands-on experience developing scalable **AI pipelines using STT and TTS technologies**. Experienced in designing **high-performance REST APIs, real-time AI workflows**, and data-driven systems for production environments. Strong understanding of **backend architecture, system integration**, and **deploying reliable, scalable AI solutions**.

TECHNICAL SKILLS

Programming : Python
AI / ML : LLMs (OpenAI, Groq), Prompt Engineering, NLP, Speech-to-Text (STT), Text-to-Speech (TTS) , LangChain, RAG
Backend : FastAPI, REST API Development, JWT Authentication, Pandas
Databases : MySQL, Redis
Tools & Platforms : Git, GitHub, GitLab, Render, Linux

PROFESSIONAL EXPERIENCE

Infosys Limited, Bangalore, India

Associate Business Analyst — Python & Gen AI | Sept 2026 – Present

- Developing and maintaining applications and backend services using Python.
- Building and integrating **Generative AI and LLM-based solutions for enterprise use cases**.
- Working on RAG (Retrieval-Augmented Generation), prompt engineering, and model orchestration.
- Developing REST APIs and backend services for AI-powered applications.
- Integrating AI models with enterprise systems and external tools.
- Working with cloud platforms including Azure, AWS, and GCP.
- Using Docker, Git, and CI/CD pipelines for application development and deployment.
- Optimizing AI solutions for scalability, performance, reliability, and response quality.
- Collaborating with cross-functional teams to understand requirements and deliver scalable technical solutions.

Deepija Telecom Private Limited, Hyderabad, India [Sept-2024 to August 2026]

Project 1: ConVox Platform Management System

- Developed core backend services including **Data Ingestion APIs, Operation Status APIs**, and **multi-endpoint workflow orchestration**. Designed and implemented **MySQL-based reporting modules** to monitor user activity, operational metrics, and system performance.
- Improved system performance by **~60%** using **Redis caching** for asynchronous tasks and high-frequency message routing.
- Enhanced platform stability through **structured testing, debugging, and API validation** using Postman.
- Collaborated closely with **product and engineering teams** to translate business requirements into **scalable backend solutions**.

Project 2: Real-Time AI Integration with ConVox

- Built a **Python-based WebSocket bridge** and **bidirectional server** to enable real-time communication with third-party AI services. Implemented an **event-driven architecture** to manage session initialization, live data streaming, and workflow completion events.
- Developed **low-latency processing modules** using Python **async programming and multithreading**.
- Integrated **webhook-based JSON reporting** to capture workflow results and generate detailed post-processing logs.
- Increased system throughput and reliability using **Redis caching, thread synchronization, retry mechanisms**, and **robust exception handling**.

Personal Project 1 : Centralized AI Automation Platform -In progress

- Developed a multi-tenant AI platform integrating Voice Bots, Chat Bots, and AI-driven Call Quality Analysis. Enabled automated customer interactions, conversation transcription, analytics, and real-time insights through a unified dashboard. Designed a scalable architecture supporting multiple clients, improving customer experience, operational efficiency, and call monitoring capabilities.
- Key Technologies:** Python, FastAPI, WebSockets, AI/LLM Integration, Speech-to-Text, Text-to-Speech, MySQL, Redis, LiveKit, Nginx, LangChain

Personal Project 2 : AI-Driven HR Attendance Management System

- Developed an **AI-enabled HR platform** for employee attendance tracking and workforce management.
- Implemented automated **analytics reports** to analyse **attendance trends and HR metrics**.
- Built an **AI assistant to answer HR-related queries** such as leave policies and portal usage.
- Ensured data **privacy controls** by restricting AI access to **sensitive employee information**.

INTERNSHIP EXPERIENCE [Deepija Telecom Pvt Ltd, Hyderabad, India | Sept 2024 – Jan 2025]

- Developed and maintained data reporting modules, processing large-scale application data using **MySQL**.
- Designed and optimized database queries for efficient data retrieval and performance improvement.
- Built and tested backend APIs using **Python** and **FastAPI** for data processing and workflow execution.
- Worked on system-level integrations, improving reliability and handling of real-time data workflows.
- Progressed into **AI-based system integration**, connecting backend services with **LLM-driven** applications for intelligent automation

CERTIFICATIONS

- IBM Prompt Engineering Basics (Coursera)
- IBM Generative AI: Introduction and Applications (Coursera)

Relevant Skills Acquired

Prompt Engineering , Large Language Models (LLMs), Generative AI , AI Applications , AI Content Generation , Prompt Design , AI Solution Development

EDUCATION :

- **Masters of Computer Applications (MCA) [2021 – 2023] |** CGPA: 8.0 / 10 | B V Raju College – Bhimavaram, Andhra Pradesh, INDIA.

STRENGTHS

- Strong problem-solving skills
- Clear understanding of AI workflows
- Quick learner in new AI tools
- Good communication & teamwork

